Huawei Access Node JAY-ACC-35 Standard Operating Procedure for Optical Transport Failure (ALARM_5115)

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_5115

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Huawei Access Node JAY-ACC-35 can occur due to the degradation of the SFP/XFP module, fiber cuts, or loss of receive/transmit power. This failure domain is specific to the Access role of the Huawei node, which relies on the integrity of the optical transport system to ensure reliable data transmission. The underlying protocol mechanics involve the Optical Transport Network (OTN) protocol, which is used to transmit data over fiber optic cables. The OTN protocol utilizes SFP/XFP modules to transmit and receive optical signals, and any degradation or failure of these modules can lead to optical transport failure. Furthermore, fiber cuts or loss of receive/transmit power can also cause optical transport failure. To ensure the reliability of the Huawei Access Node JAY-ACC-35, it is essential to regularly inspect and maintain the SFP/XFP modules, fiber optic cables, and receive/transmit power levels. This SOP outlines the steps to diagnose and resolve optical transport failure in the Huawei Access Node JAY-ACC-35.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and dispatch remote hands if necessary.

- Check the status of the redundant link to ensure it is not the primary cause of the failure.

- Gather information about the failure, including the time of occurrence, duration, and any relevant logs or alarms.

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface status, Expected Output: Interface status: UP, SFP/XFP module: OK, Description: This command displays the status of the optical transport interface, including the SFP/XFP module. If the SFP/XFP module is not OK, it may indicate a degradation or failure of the module.
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```
Command: display optical-transport alarm, Expected Output: Alarm: ALARM_5115, Optical Transport Failure, Description: This command displays any alarms related to optical transport failure, including the specific alarm code (ALARM_5115).

Command: display optical-transport interface optical power, Expected Output: Optical power: -10 dBm, Receive power: -20 dBm, Description: This command displays the optical power and receive power levels of the optical transport interface. If the optical power or receive power levels are outside the normal range, it may indicate a fiber cut or loss of receive/transmit power.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transport interface slot 1 with new SFP/XFP module, Description: Replace the SFP/XFP module with a new one to resolve the degradation or failure issue.

Step 2: Step: Verify the fiber optic cable, Command: display optical-transport interface optical power, Description: Verify that the fiber optic cable is not cut or damaged, and the optical power and receive power levels are within the normal range.

Step 3: Step: Adjust the receive/transmit power levels, Command: set optical-transport interface receive power level to -10 dBm, Description: Adjust the receive/transmit power levels to ensure they are within the normal range.

Post-Check Verification

```
Command: display optical-transport interface status, Expected Output: Interface status: UP, SFP/XFP module: OK, Description: Verify that the optical transport interface is up and the SFP/XFP module is OK after the resolution steps.

Command: display optical-transport alarm, Expected Output: Alarm: None, Optical Transport Failure, Description: Verify that there are no alarms related to optical transport failure after the resolution steps.

Command: display optical-transport interface optical power, Expected Output: Optical power: -10 dBm, Receive power: -20 dBm, Description: Verify that the optical power and receive power levels are within the normal range after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace optical-transport interface slot 1 with original SFP/XFP module, Description: Restore the original SFP/XFP module if the new one does not resolve the issue.

- Step: Verify the fiber optic cable, Command: display optical-transport interface optical power, Description: Verify that the fiber optic cable is not cut or damaged, and the optical power and receive power levels are within the normal range.

- Step: Adjust the receive/transmit power levels, Command: set optical-transport interface receive power level to original value, Description: Adjust the receive/transmit power levels to their original values if the resolution steps do not resolve the issue.